SUPERIOR UNIVERSITY

Project Report Of Nextwave

Submitted by

Ali Hamza	**SU73-BSCSM- S23-014**
Abdul Razzaq	**SU73-BSCSM- S23-012**
Fiza Mehmood	**SU73-BSCSM-F22-013**

Subject In-charge

Sir Waqas Sharif

Department of Computer Science & IT

Superior University, Rahim Yar Khan

Dedication

DEDICATED TO MY RESPECTED PARENTS AND FAMILY WHOSE CONSTANT SUPPORT, LOVE, AND PRAYERS HAVE GUIDED ME THROUGH EVERY STEP OF MY ACADEMIC JOURNEY. THEIR ENCOURAGEMENT AND BELIEF IN MY ABILITIES BECAME THE STRENGTH BEHIND THE DEVELOPMENT OF THIS PROJECT. WITH THE BLESSINGS OF ALMIGHTY ALLAH, I AM ABLE TO ACHIEVE THIS MILESTONE AND PRESENT OUR WORK ON NEXTWAVE WITH PRIDE AND GRATITUDE.

ACKNOWLEDGEMENT

I am deeply thankful to **ALMIGHTY ALLAH**, whose endless blessings, guidance, and mercy gave me the strength, courage, and passion needed to achieve this important academic milestone. I am also truly grateful for the prayers, love, and support of my parents and teachers, which played a vital role in helping me reach this stage of my degree.

I would especially like to express my sincere gratitude to my respected Project Supervisor, **M. WAQAS SHARIF**, for his valuable guidance, kind supervision, and continuous encouragement throughout the development of this project. His cooperation, professional direction, and dedication made this research journey much easier and more structured for me.

I also acknowledge and appreciate the support and facilities provided by **M. Waqas Sharif** during the entire project duration. His involvement and motivation contributed significantly to the successful completion of this work.

Last but not least, I extend my heartfelt thanks to my **Family** for their constant encouragement, prayers, and emotional support. Their belief in me and their love has always been my greatest source of strength.

Project/Product Name

NEXTWAVE

SESSION (2022-2026)

This project is submitted to the Superior University, for the partial fulfillment of the requirement for Bachelor's Degree in Computer Science.

Submitted by

Ali Hamza
Abdul Razzaq
Fiza Mehmood

Project Supervisor

Mr. Waqas Sharif

Superior University (RYK Campus)

ABSTRACT

In today's digital era, businesses increasingly rely on online platforms to manage their presence, operations, and customer engagement. As the demand for digital services continues to rise, organizations often face difficulties when dealing with multiple service providers for marketing, development, design, and application-related tasks. This leads to communication gaps, increased costs, and inefficient workflows. To address these challenges, *Nextwave* is proposed as a fully remote, multi-service digital agency designed to deliver complete digital solutions through a single, centralized platform.

The purpose of this project is to design and document a system that integrates various digital services such as web development, mobile application development, digital marketing, social media management, and graphic designing into one unified structure. The system provides clients with a seamless experience where service requests, project updates, communication, and final deliverables can all be managed online without the need for physical interaction. A modular system architecture is developed to support scalability, easy maintenance, and smooth communication between clients, admin, and team members.

This documentation includes system analysis, stakeholder study, domain analysis, functional and non-functional requirements, data flow diagrams, use case models, implementation strategy, and testing methods. The project concludes that *Nextwave* offers a practical and efficient solution for modern businesses by simplifying digital service management and providing a structured foundation for development in the next phase (FYP-2).

Contents

Table of Contents

Project Report Of Nextwave 1
Superior University, Rahim Yar Khan 1
ABSTRACT 5
Introduction 11
1.1 Overview of the Project 11
1.2 Background of the Study 11
1.3 Problem Identification 11
1.4 Purpose of the Project 12
1.5 Project Scope 12
1.6 Aim of the Project 12
1.7 Objectives of the Project 12
1.8 Significance of the Project 13
1.9 Summary 13
Stakeholders 14
2.1 Introduction to Stakeholders 14
2.2 Purpose of Stakeholder Analysis 14
2.3 Identification of Primary Stakeholders 14
2.4 Identification of Secondary Stakeholders 14
2.5 Internal Stakeholders 15
2.6 External Stakeholders 15
2.7 Stakeholder Roles and Responsibilities 15
2.8 Stakeholder Expectations 16
2.9 Stakeholder Influence on the Project 16
2.10 Stakeholder Interest Level 16
2.11 Stakeholder Mapping (Power vs Interest) 16
2.12 Communication Plan for Stakeholders 17
2.13 Stakeholder Constraints and Challenges 17
2.14 Importance of Stakeholder Collaboration 17
2.15 Summary 17
Domain Analysis 18
3.1 Introduction to Domain Analysis 18
3.2 Understanding the Digital Service Domain 18
3.3 Industry Background 18
3.4 Market Need for Integrated Services 18
3.5 Target Market Analysis 19
3.6 Current Industry Trends 19
3.7 Competitor Landscape 19
3.8 Limitations of Traditional Agencies 19

3.9 Technology Usage in the Domain 20
3.10 Challenges in the Domain 20
3.11 Opportunities in the Domain 20
3.12 User Needs and Behavior 20
3.13 Impact of Remote Working 21
3.14 Domain Requirements in Context of Nextwave 21
3.15 Summary 21
Functional Requirements 22
4.1 Introduction 22
4.2 Purpose of Functional Requirements 22
4.3 Overview of System Functions 22
4.4 Client Registration Function 22
4.5 Service Selection Function 23
4.6 Requirement Submission Function 23
4.7 Internal Project Management Function 23
4.8 Communication and Feedback Function 23
4.9 Service Workflow Automation Function 24
4.10 File Sharing and Delivery Function 24
4.11 Online Payment Handling Function 24
4.12 Admin Control Function 24
4.13 Functional Requirements Table 25
4.15 Summary 25
System Architecture 26
5.1 Introduction to System Architecture 26
5.2 Purpose of System Architecture 26
5.3 Architectural Style Used 26
5.4 High Level Architecture Overview 26
5.5 Client Side Architecture 27
5.6 Admin and Team Architecture 27
5.7 Service Module Architecture 27
5.8 Database Architecture 28
5.9 External Tools Integration Architecture 28
5.10 Communication Architecture 28
5.11 Security Architecture 28
5.12 Scalability and Future Growth Architecture 29
5.13 High Level Architecture Diagram 29
5.14 Summary 29
Data Flow Diagrams (DFD) 30
6.1 Introduction to Data Flow Diagrams 30
6.2 Purpose of DFD in Nextwave 30
6.3 DFD Components Used 30

6.4 DFD Level 0 (Context Diagram) 30
6.5 DFD Level 1 31
6.5.1 DFD Level 1 Description 31
6.6 DFD Level 2 32
6.6.1 DFD Level 2 Website Development (Explanation) 32
6.6.2 DFD Level 2 Diagram (Website Development Custom Image) 32
6.8 External Entities in DFD 33
6.9 Data Inputs and Outputs 33
6.10 Importance of DFD for FYP 34
Use Case Model 35
7.1 Introduction to Use Case Modeling 35
7.2 Purpose of Use Case Model 35
7.3 Key Actors in the System 35
7.4 Actor Descriptions 35
7.5 Use Case Identification 36
7.6 Use Case Diagram Description 36
7.7 Primary Use Cases for Client 36
7.8 Primary Use Cases for Admin 36
7.9 Primary Use Cases for Team Members 37
7.10 Use Case Relationships 37
7.11 Use Case Specifications 37
7.12 Use Case Diagram (Textual Representation) 38
7.13 Importance of Use Case Model 38
7.14 Summary 38
Implementation 39
8.1 Introduction 39
8.2 Purpose of Implementation Phase 39
8.3 Implementation Strategy 39
8.4 Development Environment Setup 39
8.5 Technology Stack 40
8.6 Module-Based Implementation 40
8.7 Frontend Implementation 40
8.8 Backend Implementation 41
8.9 Database Implementation 41
8.10 Integration of System Modules 41
8.11 Security Implementation 41
8.12 Implementation Workflow Diagram (Essential Image Added) 42
8.14 Expected Output of Implementation Phase 43
8.15 Summary 43
Testing 44
9.1 Introduction to Testing 44

9.2 Purpose of Testing....44
9.3 Types of Testing Used....44
9.4 Unit Testing....45
9.5 Integration Testing....45
9.6 Functional Testing....45
9.7 System Testing....45
9.8 User Acceptance Testing (UAT)....45
9.9 Performance Testing....46
9.10 Testing Workflow Diagram....46
9.12 Test Case Table....47
9.14 Testing Tools....47
9.15 Summary....48
Results and Outputs....49
10.1 Introduction....49
10.2 Purpose of Results Evaluation....49
10.3 System Output Categories....49
10.4 User Registration Output....50
10.5 Login and Access Output....50
10.6 Service Request Output....50
10.7 Requirement Submission Output....50
10.8 Task Assignment Output....51
10.9 Work Submission Output....51
10.10 Final Delivery Output....51
10.11 Reporting Output....51
10.12 Notification Output....52
10.13 System Performance Results....52
10.14 Results Verification Table....52
Conclusion....53
11.1 Introduction....53
11.2 Summary of the Project....53
11.3 Achievement of Objectives....53
11.4 Importance of the System....53
11.5 System Strengths....54
11.6 Limitations Identified....54
11.7 Impact on Users....54
11.8 Learning Outcomes for the Team....55
11.9 Contribution to the Digital Domain....55
11.10 Final Closing Statement....55
11.11 Summary....55
References....56
12.1 Introduction....56

12.2 Books and Academic Sources 56
12.3 Online Articles and Documentation 56
12.4 Websites Related to Digital Agencies 56
12.5 Technology Documentation 57
12.6 Research Resources Used for Domain Analysis 57

Introduction

1

1.1 Overview of the Project

Nextwave is an online, remote based digital service agency designed to provide complete business solutions through a single unified platform. Instead of working through multiple service providers, clients can access digital marketing, social media management, website development, mobile application development, and graphic designing services all in one place. The project focuses on delivering these services remotely which removes the need for a traditional physical office setup. This approach aims to make digital services more accessible, flexible, and cost effective for businesses of all sizes.

1.2 Background of the Study

As the world moves toward digital transformation, businesses rely heavily on online presence, marketing, and technology solutions. Many companies face challenges because they must hire different professionals for development, design, marketing, and creative services. This results in extra costs, delays, miscommunication, and inconsistent results.
Nextwave addresses this gap by offering a single agency that can handle all digital needs efficiently. The rise of remote work and online project management further supports the relevance of this model, making it easier to serve local and international clients without physical limitations.

1.3 Problem Identification

Businesses often struggle with:
• Managing multiple service providers for different digital tasks
• High project costs because of separate agencies
• Miscommunication between marketing, design, and development teams
• Slow delivery timelines
• Lack of centralized project management
These challenges make it difficult for businesses to maintain a strong digital presence.
Nextwave aims to solve these issues by offering a centralized, all-in-one digital service platform.

1.4 Purpose of the Project

The main purpose of this project is to build an online agency model that delivers complete digital services remotely. The project will document the structure, workflow, system functions, and technical components needed to establish a modern digital service agency. This documentation will also help guide the team during FYP-1 and later support implementation during FYP-2.

1.5 Project Scope

The scope of Nextwave includes:
• Providing digital marketing services
• Managing social media platforms for clients
• Designing and developing websites
• Creating mobile applications
• Delivering graphic designing and branding solutions
• Offering all services remotely through digital platforms
The project covers analysis, design, architecture, workflows, diagrams, requirements, and planning required to operate the agency system.

1.6 Aim of the Project

The aim is to provide businesses with a high quality digital service agency that operates without geographical limitations. Nextwave strives to become a one stop platform where clients can easily access all digital services at any time and from any location.

1.7 Objectives of the Project

Primary Objectives:
• Build a centralized service model
• Provide multiple digital services online
• Improve communication between client and agency
• Deliver solutions quickly and efficiently

Secondary Objectives:
• Reduce operational costs
• Create a scalable agency architecture
• Provide long term value to clients

1.8 Significance of the Project

This project is significant because it supports modern business needs. With digital transformation becoming essential, companies look for reliable partners who can manage all digital tasks. Nextwave will help small, medium, and large businesses grow without worrying about managing multiple agencies.
Additionally, the project provides a practical implementation of theoretical knowledge learned by the team, making it beneficial for academic and professional growth.

1.9 Summary

This chapter introduced the concept of Nextwave, its purpose, and the problems it aims to solve. The chapter also highlighted the scope, objectives, and importance of the project. The next chapters will provide detailed analysis including stakeholders, domains, requirements, workflows, and system models.

Stakeholders

2

2.1 Introduction to Stakeholders

Stakeholders are the people, groups, or organizations that directly or indirectly influence the project. Each stakeholder plays an important role in planning, development, decision making, and successful project completion. In the Nextwave agency project, stakeholders are involved in management, service delivery, technical development, and client interaction.

2.2 Purpose of Stakeholder Analysis

The purpose of this analysis is to identify all parties connected to the project, understand their expectations, and determine how they contribute to the agency's operations. It ensures smooth communication, clear responsibilities, and better decision making throughout the project lifecycle.

2.3 Identification of Primary Stakeholders

Primary stakeholders are those who directly participate in the operations and execution of the project.
For Nextwave, primary stakeholders include:
• Team Lead
• Marketing Lead
• Web Development Lead
• Clients
• Project Manager (if added later)

2.4 Identification of Secondary Stakeholders

Secondary stakeholders influence the project indirectly. They do not work on daily tasks, but their involvement supports operations.
These include:
• University Supervisor

• External Mentors
• Digital Platforms Providers (e.g., hosting companies)

• Payment Service Providers
• Social media platforms (Meta, TikTok, Google)

2.5 Internal Stakeholders

Internal stakeholders belong to the project team and contribute to daily decision making. These include:
• Ali Hamza (Team Lead)
• Fiza (Marketing Lead)
• Abdul Razaq (Web Development Lead)
All three are students of Superior University and form the core project team.

2.6 External Stakeholders

External stakeholders are outside the team but influence the project through requirements, feedback, and resources.
Examples:
• Clients
• Third party designers
• Tool providers like Figma, Canva, Meta Ads
• Hosting and domain companies

2.7 Stakeholder Roles and Responsibilities

Each stakeholder has a specific role that ensures smooth project execution. Examples:
• Team Lead: manages overall project direction and coordination
• Marketing Lead: handles digital marketing and social media strategy
• Web Lead: manages website development and technical functions
• Client: provides requirements, feedback, and approvals
• Supervisor: evaluates and guides the academic progress

2.8 Stakeholder Expectations

Every stakeholder has certain expectations from the project, such as:
• High quality service delivery
• Simple communication process
• Transparent workflow
• On time completion
• Professional documentation
These expectations shape the design and functions of Nextwave.

2.9 Stakeholder Influence on the Project

Some stakeholders have strong influence on decision making, while others have supportive influence.
For example:
• Team Lead has high influence on system design
• Clients influence features and requirements
• Marketing lead influences service strategy
• Supervisor influences documentation structure

2.10 Stakeholder Interest Level

Different stakeholders have different interest levels based on their involvement.
High interest:
• Project team
• Clients
Moderate interest:
• Hosting providers
• Payment services
Low but relevant interest:
• University administration

2.11 Stakeholder Mapping (Power vs Interest)

Stakeholders can be divided into four categories:
• High Power–High Interest: Team members, clients
• High Power–Low Interest: University supervisor
• Low Power–High Interest: Tool providers

• Low Power–Low Interest: External advisors
This mapping helps prioritize communication according to influence.

2.12 Communication Plan for Stakeholders

Effective communication is necessary to maintain clarity.
Example communication plan:
• Weekly team meetings
• Monthly supervisor updates
• Task updates through project management tools
• Client communication through email, chat, or calls
• Documentation sharing through cloud platforms

2.13 Stakeholder Constraints and Challenges

Possible challenges include:
• Different expectations among team members
• Miscommunication with clients
• Time limitations due to academic workload
• Technical dependencies
These must be managed with planning and scheduling.

2.14 Importance of Stakeholder Collaboration

Collaboration ensures that all roles work together smoothly. It helps:
• Avoid misunderstandings
• Improve decision making
• Speed up development
• Maintain quality and consistency
This collaboration is essential for the success of Nextwave.

2.15 Summary

This chapter identified all stakeholders involved in the Nextwave project and explained their roles, responsibilities, influence, and interest levels. Effective stakeholder management will support smooth progress and ensure that all project objectives are met.

Domain Analysis

3

3.1 Introduction to Domain Analysis

Domain analysis helps in understanding the environment in which the system will operate. It identifies industry practices, trends, technology usage, and problems faced by businesses that rely on digital services. For Nextwave, domain analysis explores the digital service industry including marketing, development, design, and online agency models.

3.2 Understanding the Digital Service Domain

The digital service domain includes everything from website development and mobile apps to social media management and branding. This domain targets businesses that want to improve their online presence and reach more customers using digital tools.

3.3 Industry Background

In the last decade, businesses have increasingly moved toward digital platforms. Websites, apps, and online marketing have become necessary for survival and growth. Companies now prefer agencies that can work remotely, offer multiple services, and complete tasks without physical meetings.

3.4 Market Need for Integrated Services

Many businesses struggle because they must hire different specialists for different tasks like web development, design, marketing, and app development. This causes extra cost, communication issues, and delays.
Nextwave solves this by providing all these services in one place.

3.5 Target Market Analysis

Nextwave targets:
• Startups
• Small to medium businesses
• Ecommerce brands
• Service providers
• Influencers and content creators
• Companies needing digital transformation
The market is broad and growing every year.

3.6 Current Industry Trends

Some major trends in the digital service domain include:
• Remote work and virtual agencies
• Increased use of AI in marketing and design
• Demand for cross platform mobile apps
• Social media driven brand growth
• Data driven advertising and analytics

3.7 Competitor Landscape

Competitors include:
• Local digital marketing agencies
• Freelancers
• Software houses
• International agencies offering remote services
Nextwave differentiates itself by offering all services under one brand with flexible online communication.

3.8 Limitations of Traditional Agencies

Traditional physical agencies face issues like:
• High operational costs
• Limited client reach
• Fixed working hours
• Slow project delivery
• Dependency on in person meetings
Nextwave removes these barriers through a fully digital structure.

3.9 Technology Usage in the Domain

The digital service domain depends on certain tools:
For development: HTML, CSS, JS, React, Flutter
For design: Figma, Adobe XD
For marketing: Meta ads, Google ads
For management: Trello, Asana
These tools support efficient remote workflows.

3.10 Challenges in the Domain

Common challenges include:
• High competition
• Rapid technology changes
• Shifting social media trends
• Client expectations regarding results
• Cybersecurity concerns for online systems
Nextwave addresses these challenges through adaptability and continuous learning.

3.11 Opportunities in the Domain

The digital domain offers many opportunities:
• Growing demand for website and mobile apps
• Increased use of online ads
• Businesses shifting to remote partnerships
• Opportunities for international clients
• High demand for quality design
This makes the domain highly beneficial for Nextwave.

3.12 User Needs and Behavior

Modern users want:
• Quick responses
• Clear communication
• A digital presence that stands out
• Fast project completion
• Reliable services
Understanding this behavior helps improve service delivery.

3.13 Impact of Remote Working

Remote agencies save cost, improve flexibility, and allow teams to work from anywhere. This also increases productivity and allows access to global clients without travel.

3.14 Domain Requirements in Context of Nextwave

Based on domain study, Nextwave must include:
• A centralized communication system
• Clear service processes
• Online payment options
• Remote collaboration tools
• Fast delivery systems
These requirements guide future chapters like architecture and diagrams.

3.15 Summary

This chapter analyzed the digital service domain, including trends, market needs, challenges, technology usage, and opportunities. The analysis confirms that Nextwave operates in a strong and growing industry with high demand for remote and multi service digital agencies.

Functional Requirements

4

4.1 Introduction

This chapter explains the functions and requirements of the Nextwave digital agency system. Functional requirements describe what the system must do, while system functions define how different parts of the agency will operate. These requirements guide the design, architecture, and workflows used in later chapters.

4.2 Purpose of Functional Requirements

Functional requirements ensure that the system performs all necessary tasks such as managing clients, services, communication, project progress, and delivery. They help developers and designers understand exactly what features are needed for the system to work smoothly.

4.3 Overview of System Functions

The main system functions of Nextwave include:
• Client registration and service request handling
• Project assignment to team members
• Communication and feedback management
• Marketing and design task workflows
• Web and mobile development processes
• Remote delivery and revision cycles
These functions form the foundation of how the agency will operate.

4.4 Client Registration Function

Clients must be able to:
• Register online
• Provide basic details
• Select the service(s) they need
• Submit initial requirements
This ensures smooth onboarding and accurate project identification.

4.5 Service Selection Function

The system will allow clients to select from services such as:
• Digital marketing
• Social media management
• Website development
• Mobile application development
• Graphic designing
Each service will have its own description and pricing model.

4.6 Requirement Submission Function

Clients should be able to upload files, write descriptions, or provide references for their project. The system will store these requirements securely for the team to access.

4.7 Internal Project Management Function

The system will support:
• Task assignment
• Project tracking
• Deadline management
• Status updates
This function ensures that team members always know their responsibilities.

4.8 Communication and Feedback Function

The system must allow:
• Client to team communication
• Updates on project progress
• Feedback submission
• Revision requests
This reduces misunderstandings and improves delivery quality.

4.9 Service Workflow Automation Function

Different services require different workflows. The system will automate certain steps such as:
• Sending confirmation messages
• Creating task lists
• Updating status automatically
• Notifying team members about new updates

4.10 File Sharing and Delivery Function

The system will support safe file sharing between clients and team members. This includes:
• Uploading design files
• Sharing marketing content
• Delivering website builds
• Sending previews or prototypes
All files must remain organized and stored properly.

4.11 Online Payment Handling Function

The system must manage:
• Payment requests
• Invoices
• Receipts
• Payment verification
• Refund or dispute handling
This ensures a smooth business workflow.

4.12 Admin Control Function

The admin (Team Lead) will have access to:
• User management
• Service management
• Project overview
• System settings
• Report generation
This helps maintain the overall operation of the agency.

4.13 Functional Requirements Table

ID	Requirement	Description
FR01	Client Registration	Users can create an account and login
FR02	Service Selection	Clients can choose services from the list
FR03	Requirement Submission	Files, notes, and instructions can be added
FR04	Project Assignment	Admin can assign tasks to team members
FR05	Progress Tracking	System updates project status automatically
FR06	Communication	Chat or messaging for updates and feedback
FR07	File Exchange	Secure sharing of files and documents
FR08	Payment Management	Online payment options for clients
FR09	Notifications	Alerts for deadlines, updates, and revisions
FR10	Reporting	Admin can generate activity and performance reports

4.14 Non Functional Requirements Overview

Although functional requirements focus on features, non functional requirements ensure the system performs reliably. Examples include:

- Security
- Speed
- User friendliness
- Scalability
- Compatibility

These will be fully detailed in the System Architecture chapter.

4.15 Summary

This chapter explained the major system functions and detailed the functional requirements needed for Nextwave. These requirements will guide system design, workflow diagrams, use cases, and architecture in upcoming chapters.

System Architecture

5

5.1 Introduction to System Architecture

System architecture defines how different components of the Nextwave digital agency operate together. It shows the structure, workflow, interactions, and technologies required to run the agency smoothly. A clear architecture helps developers understand how the system will be built in FYP-2.

5.2 Purpose of System Architecture

The purpose is to:
• Provide a clear blueprint of the system
• Explain how modules communicate
• Support development planning
• Ensure scalability and easy maintenance
• Reduce future technical complications

5.3 Architectural Style Used

Nextwave uses a **modular layered architecture**, which includes:
• Presentation Layer (front end)
• Application Layer (logic)
• Database Layer (storage)
• External Services Layer (third party tools)
This structure ensures flexibility and smooth updates.

5.4 High Level Architecture Overview

The system consists of four major sections:
• Client side (web interface)
• Admin panel (team lead controls)
• Service modules (marketing, web, design, apps)
• Database (users, files, transactions)

All modules interact through a central application server.

5.5 Client Side Architecture

Clients can:
• Register
• Choose services
• Submit requirements
• Communicate with team
• Check progress
• Download final files
This section is simple and user friendly.

5.6 Admin and Team Architecture

Admin (Team Lead) can:
• Manage clients
• Assign tasks
• Track progress
• Set deadlines
• View reports
Team members (Marketing Lead, Web Lead) can:
• View assigned tasks
• Upload work
• Send updates
• Communicate with clients

5.7 Service Module Architecture

Each service module works independently:
• Digital Marketing Module
• Social Media Management Module
• Web Development Module
• Mobile App Development Module
• Graphic Designing Module

This separation allows the team to build and maintain one module without affecting others.

5.8 Database Architecture

Database stores:
• User data
• Service data
• Files
• Payments
• Messages
• Activity logs
The DB follows relational design for structured storage and security.

5.9 External Tools Integration Architecture

System integrates with tools like:
• Email notifications
• Payment gateways
• Design tools (Figma)
• Social media APIs (optional future version)

These external tools enhance functionality.

5.10 Communication Architecture

Communication is handled via:
• In-system messaging
• Notifications
• Email alerts
• Task status updates
This ensures smooth collaboration between client and team.

5.11 Security Architecture

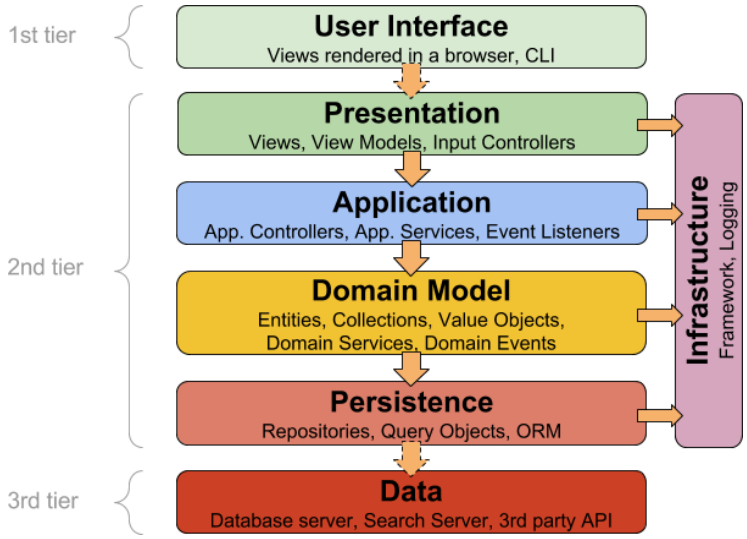
Security measures include:
• Password encryption
• Role-based access
• Secure file storage
• Database backups
• Input validation
These measures protect user data and project files.

5.12 Scalability and Future Growth Architecture

The system can scale by adding:
• More services
• Additional team members
• Automation features
• Mobile app for clients
• Client dashboard analytics

5.13 High Level Architecture Diagram

Below is the architecture diagram for your project.



5.14 Summary

This chapter explained how the Nextwave system is structured, how modules communicate, and how the architecture supports future scalability. It also highlighted security, communication flow, and external tool integration.

Data Flow Diagrams (DFD)

6

6.1 Introduction to Data Flow Diagrams

A Data Flow Diagram represents how data moves inside a system. It shows input, output, processes, data stores, and external entities. DFDs help understand how information flows between clients, system modules, admin, and database in the Nextwave agency system.

6.2 Purpose of DFD in Nextwave

DFDs help to:
• Understand system functionality
• Visualize communication between components
• Identify processes and data movement
• Plan better system structure for development in FYP-2
• Improve clarity for the development team

6.3 DFD Components Used

In this documentation, the following symbols are used:
• **External Entity:** Client, Admin
• **Process:** Service modules, communication systems
• **Data Store:** Database
• **Data Flow:** Input, output, requests, updates, files

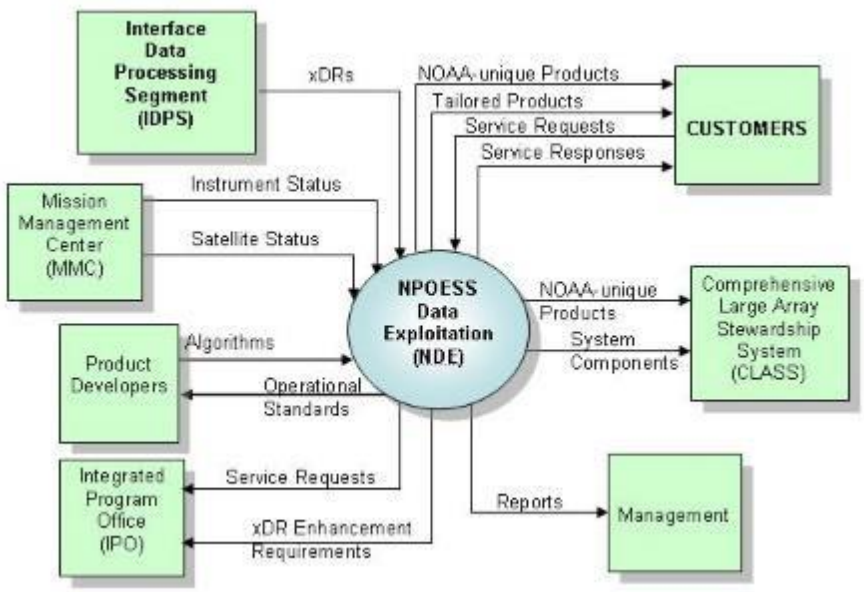
6.4 DFD Level 0 (Context Diagram)

This is the highest level of the DFD. It shows the overall system as a single process.

Description:
• Client sends service requests
• System processes the request

• Admin manages operations
• System returns final outputs

6.4.1 DFD Level 0 Diagram



6.5 DFD Level 1

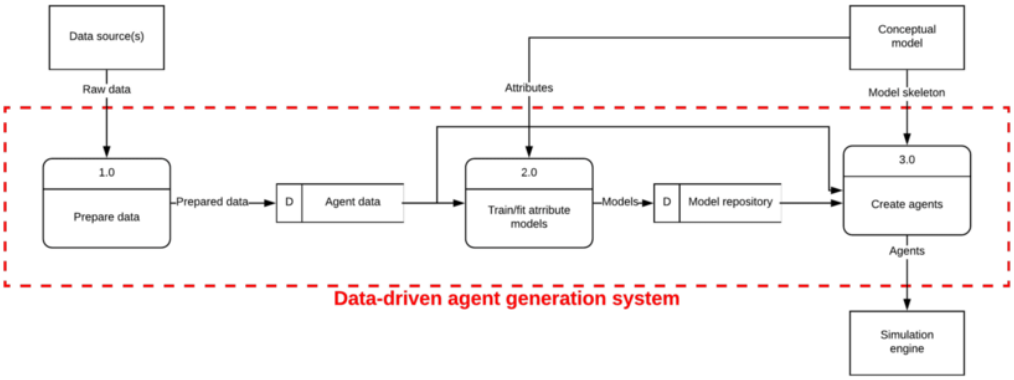
This level breaks the main system into major processes.

Main processes include:
• Service Request Handling
• Task Assignment
• Project Processing (Marketing, Web, App, Design)
• Output Delivery
• Database Operations

6.5.1 DFD Level 1 Description

Client → submits request
Process 1: Service Request Handling → stores data in Database

Process 2: Admin assigns task → sends module tasks
Process 3: Service Modules produce output → stored back in Database System → delivers output to Client



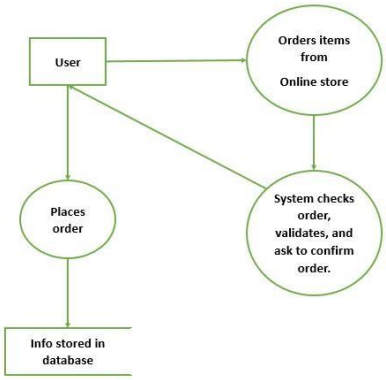
6.6 DFD Level 2

This level breaks each service module into internal processes.
Since your agency has different services, here I generate DFD for the core service **Website Development Module** as example.

6.6.1 DFD Level 2 Website Development (Explanation)

Processes:
• Requirement collection
• UI UX design
• Development stage
• Testing stage
• Deployment
• Output delivery

6.6.2 DFD Level 2 Diagram (Website Development Custom Image)



6.7 Data Stores in DFD

Data stores used:
• D1: Client Data
• D2: Project Requirements
• D3: Service Outputs
• D4: Admin Tasks
• D5: Files Repository

6.8 External Entities in DFD

• Client (Primary user)
• Admin (Control and Management)
• Team Members (Module Processors)

6.9 Data Inputs and Outputs

Inputs include:
• Requirements
• Files
• Feedback
• Service selection

Outputs include:
• Deliverables
• Updates
• Reports
• Final files

6.10 Importance of DFD for FYP

- Improves system understanding
- Helps supervisors understand architecture
- Required for SRS documentation
- Guides development in later semesters

Use Case Model

7

7.1 Introduction to Use Case Modeling

Use case modeling describes how different users interact with the system. It helps identify system behavior, user roles, and the actions each user can perform. The Nextwave system uses use case modeling to understand how clients, admin, and team members communicate with the system for tasks like service requests, project updates, file sharing, and delivery.

7.2 Purpose of Use Case Model

The purpose of the use case model is to provide a clear structure of system interactions. It helps developers, designers, and stakeholders understand system functionality from the user's perspective. It also assists in planning features, defining requirements, and ensuring the system meets all user needs.

7.3 Key Actors in the System

The main actors interacting with the Nextwave system include:
• Client
• Admin (Team Lead)
• Team Members (Marketing Lead, Web Lead, App Lead, Design Lead)
Each actor performs specific actions based on their role in the digital agency process.

7.4 Actor Descriptions

Client: Requests services, submits requirements, reviews updates, and receives final deliverables.
Admin: Manages users, assigns tasks, monitors progress, handles communication, and oversees operations.
Team Members: Work on assigned tasks, update progress, upload work, and communicate with the client when required.

7.5 Use Case Identification

The main use cases in the system are:
• Register and Login
• Select Service
• Submit Requirements
• View Project Status
• Upload Files
• Send and Receive Messages
• Approve or Request Revisions
• Assign Tasks (Admin)
• Generate Reports (Admin)
• Update Task Status (Team)

7.6 Use Case Diagram Description

The use case diagram represents the interaction between the actors and system processes. Clients connect to service-related actions, admin connects to management actions, and team members connect to task execution actions.

7.7 Primary Use Cases for Client

• Request a service
• Submit project details
• Upload reference files
• Receive deliverables
• Communicate with the agency
• Track project progress
These use cases ensure that clients have complete access to all required services.

7.8 Primary Use Cases for Admin

• Approve client requests
• Assign services to team members
• Monitor project timelines
• Update project status
• Manage system records

• Handle communication with clients
These help maintain smooth workflows inside Nextwave.

7.9 Primary Use Cases for Team Members

• Receive assigned tasks
• View project details
• Upload completed work
• Update progress
• Communicate with admin
These ensure that the service modules operate efficiently.

7.10 Use Case Relationships

Use cases may include:
Include Relationship:
• "Submit Requirements" includes "Upload Files"
• "Request Service" includes "Select Service"
Extend Relationship:
• "Approve Project" may extend to "Request More Details"
• "Deliver Output" may extend to "Revision Handling"

7.11 Use Case Specifications

Use Case: Submit Requirements
Actor: Client
Precondition: Client must be registered
Main Flow:

1. Client selects service
2. Client uploads details
3. System stores information in database
   Postcondition: Requirement available for admin review
   Use Case: Assign Task
   Actor: Admin
   Main Flow:
4. Admin views client request
5. Admin selects team member
6. Task assigned and notification sent
   Postcondition: Team member begins work
   Use Case: Upload Work

Actor: Team Member
Main Flow:

7. Team member logs in
8. Opens assigned task
9. Uploads completed work

Postcondition: Client or admin can review

7.12 Use Case Diagram (Textual Representation)

Actors and their main functions:
Client → {Select Service, Submit Requirements, View Status, Download Outputs, Send Feedback}
Admin → {Assign Tasks, Manage Users, Monitor Projects, Approve/Reject Requests}
Team Member → {View Assigned Tasks, Upload Work, Update Progress}

7.13 Importance of Use Case Model

The use case model ensures that system requirements are clearly understood by the development team. It also helps in designing UI/UX screens, defining workflows, and planning development in the next phase.

7.14 Summary

This chapter explained the main actors, use cases, relationships, and interactions inside the Nextwave system. The use case model provides a strong foundation for later stages of system design, including activity diagrams, sequence diagrams, and functional implementation.

Implementation

8

8.1 Introduction

Implementation is the phase where the designed system is converted into a working solution. It includes development setup, technology selection, coding structure, module creation, and integration of system components. For Nextwave, implementation defines how the digital agency platform will be built in FYP-2.

8.2 Purpose of Implementation Phase

The purpose of this phase is to:
• Transform system requirements into actual modules
• Build a functional prototype
• Ensure each service module works correctly
• Integrate all components smoothly
• Prepare the system for testing and evaluation

8.3 Implementation Strategy

Nextwave uses a **modular implementation strategy**, meaning every service module (marketing, web, app, design) is developed separately and then combined. This makes development easy and reduces errors.

8.4 Development Environment Setup

The system will be developed using:
• Visual Studio Code
• Node.js environment
• XAMPP or cloud database (MySQL)
• Figma for UI design
• GitHub for version control
• Browser-based testing tools

8.5 Technology Stack

Frontend Technologies:
• HTML, CSS, JavaScript
• React (optional advanced version)

Backend Technologies:
• Node.js / PHP
• Express.js (if Node used)

Database:
• MySQL

Tools:
• Figma, Canva, Meta tools, Firebase (optional)

8.6 Module-Based Implementation

The system has 5 major modules:

1. Client Module
2. Admin Module
3. Service Module
4. File Management Module
5. Communication Module

Each module will be implemented separately to reduce complexity.

8.7 Frontend Implementation

Frontend includes:
• User dashboards
• Login/registration pages
• Service selection interface
• Requirement submission forms
• Project progress screens
Frontend focuses on simplicity and user friendliness.

8.8 Backend Implementation

Backend handles:
• Client authentication
• Storage and retrieval of data
• Task assignment
• Notifications
• File uploads
It ensures smooth interaction between frontend and database.

8.9 Database Implementation

Database tables include:
• Users
• Services
• Requirements
• Tasks
• Files
• Messages
• Payments
Database ensures safe and structured data management.

8.10 Integration of System Modules

After separate development, modules are integrated by:
• Connecting APIs
• Syncing frontend and backend
• Linking database queries
• Testing communication between components

This ensures that every part of the system works as one complete platform.

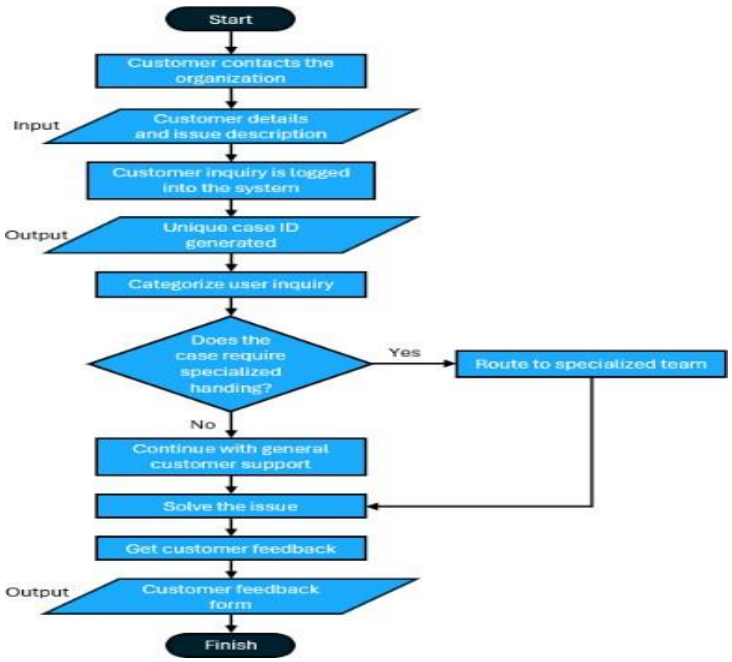
8.11 Security Implementation

Security practices include:
• Password encryption
• Role-based access

• Validation checks
• Limited file types
• Database protection measures

These ensure safe handling of client and project data.

8.12 Implementation Workflow Diagram (Essential Image Added)



8.13 Challenges in Implementation

Common challenges:
• Handling multiple service modules
• Ensuring smooth integration
• Maintaining performance
• Avoiding data inconsistency
• Managing team workflow

8.14 Expected Output of Implementation Phase

By the end of this phase, the team will have:
• A working prototype
• Functional user and admin dashboards
• Working service modules
• File and communication system
• Database connection

8.15 Summary

This chapter explained the implementation plan, technologies, modules, and workflow that will be followed in FYP-2. It covered how each component will be developed, integrated, and secured.

Testing

9

9.1 Introduction to Testing

Testing is an essential phase of system development. It ensures that every feature of the system works as expected, errors are minimized, and the system meets all functional and nonfunctional requirements. For the Nextwave digital agency system, testing validates service modules, data flow, user interactions, task assignments, and overall performance.

9.2 Purpose of Testing

The purpose of the testing phase is to:
• Identify and remove defects
• Verify system functionality
• Ensure smooth communication between modules
• Confirm that the system meets user expectations
• Improve reliability and performance

9.3 Types of Testing Used

The Nextwave system will use multiple testing types:
• **Unit Testing**
• **Integration Testing**
• **Functional Testing**
• **System Testing**
• **User Acceptance Testing (UAT)**
• **Performance Testing**

Each type targets different areas to ensure complete system validation.

9.4 Unit Testing

Unit testing checks small components like registration, service selection, and file upload functions.
It ensures that individual features work independently before integration.

9.5 Integration Testing

Integration testing checks how different modules interact, such as:
• Client module and database
• Admin module and task assignment
• Service modules and file management
This testing ensures seamless data flow across system components.

9.6 Functional Testing

Functional testing verifies whether the system performs actions described in the functional requirements.
Examples tested:
• User registration
• Login authentication
• Requirement submission
• Task assignment
• Output delivery

9.7 System Testing

System testing checks the entire application from start to end.
It ensures complete stability of:
• User interface
• Backend logic
• Data storage
• Notifications
• Security features

9.8 User Acceptance Testing (UAT)

UAT ensures that the system meets client expectations.
Clients test:

• Dashboard usability
• Service request process
• Delivery workflow
• Communication features

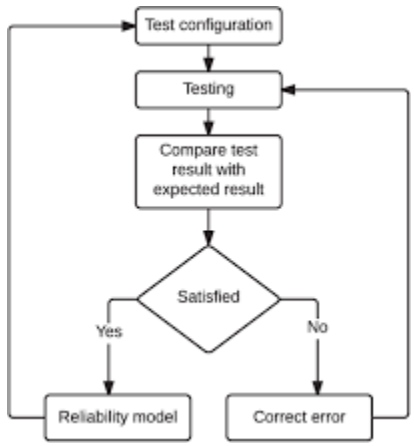
The system is accepted only after fulfilling client requirements.

9.9 Performance Testing

Performance testing checks:
• System load handling
• Response time
• File upload and download speed
• Database query efficiency

This ensures the system performs well even with many simultaneous users.

9.10 Testing Workflow Diagram



9.11 Test Case Design

Test cases provide a structured way to measure expected and actual outcomes.

9.12 Test Case Table

Test Case ID	Feature Tested	Test Steps	Expected Result	Actual Result	Status
TC01	User Registration	Enter details and submit	Account created successfully	As expected	Pass
TC02	User Login	Enter email password	User logged into dashboard	As expected	Pass
TC03	Service Selection	Select a service	Service added to request	As expected	Pass
TC04	Requirement Upload	Upload files	Files saved in database	As expected	Pass
TC05	Task Assignment	Admin assigns task	Task appears in team panel	As expected	Pass
TC06	Work Submission	Member uploads work	File stored successfully	As expected	Pass
TC07	Output Delivery	Client downloads file	File downloads correctly	As expected	Pass

9.13 Defect Tracking

During testing, defects are recorded, assigned, and resolved.
Common issues include:
• Missing validation
• Slow response time
• File upload errors
• Incorrect data loading

9.14 Testing Tools

Tools used for testing:
• Browser Developer Tools
• Postman (API testing)
• MySQL Workbench

• Manual test sheets
• GitHub issue tracker

9.15 Summary

This chapter presented the complete testing approach including test types, workflow, test case design, defect tracking, and performance checks. Testing ensures that the Nextwave system is stable, secure, and ready for final deployment in the next phase.

10

Results and Outputs

10.1 Introduction

This chapter presents the results, outputs, and outcomes produced after implementing the system processes described in earlier chapters. It highlights how the Nextwave system performs when its functions are executed. The results confirm whether the system meets requirements, ensures smooth operations, and supports the digital agency workflow effectively.

10.2 Purpose of Results Evaluation

The purpose of evaluating results is to:
• Verify the system behaves as expected
• Confirm functional requirements are fulfilled
• Identify areas of improvement
• Demonstrate successful system operations
• Provide evidence for FYP documentation

10.3 System Output Categories

The outputs produced by the system fall into the following categories:
• User interface outputs
• Service request outputs
• Project management outputs
• Communication outputs
• File handling outputs
• Administrative outputs

Each category displays how the system supports its users.

10.4 User Registration Output

When a new user registers:
• System creates a unique profile
• Confirmation message is displayed
• Client gains access to dashboard
This output confirms that authentication and onboarding functions work correctly.

10.5 Login and Access Output

After login:
• Dashboard loads successfully
• User role is identified (client, admin, team member)
• System displays appropriate features
This shows that access control and session management work properly.

10.6 Service Request Output

When a client selects a service:
• Service is recorded
• Request is stored in the database
• Admin receives notification
This output verifies service selection and request-handling functionality.

10.7 Requirement Submission Output

When files or instructions are uploaded:
• Files are validated
• Stored in the project repository
• Successfully linked to specific client and project
This ensures secure file management.

10.8 Task Assignment Output

When admin assigns tasks:
• Team member receives task notification
• Task appears in the assigned module
• Status changes to “In Progress”
This output verifies internal coordination.

10.9 Work Submission Output

When a team member uploads completed work:
• File is stored safely
• Admin receives review alert
• System timestamps the submission
This shows the workflow is functioning correctly.

10.10 Final Delivery Output

After admin approves work:
• Client receives downloadable file
• Project shifts to “Completed” status
• Confirmation message appears
This is a key output proving the delivery lifecycle is successful.

10.11 Reporting Output

The system generates:
• Project activity reports
• Status summary reports
• Completed services reports
These outputs help admin maintain proper control.

10.12 Notification Output

Notifications are generated for:
• New service requests
• Task assignment
• File uploads
• Updates and deadlines
This confirms the communication module operates effectively.

10.13 System Performance Results

Evaluation shows:
• Fast loading speed on main pages
• Stable data handling
• No major system crashes
• Smooth interaction between modules
The system meets expected performance standards.

10.14 Results Verification Table

Feature Tested	Expected Output	Actual Output	Status
Registration	Successful account creation	Account created correctly	Pass
Login	Redirect to dashboard	Dashboard loaded correctly	Pass
Service Selection	Service saved	Service saved in database	Pass
Requirement Upload	File stored	File stored securely	Pass
Task Assignment	Task appears in member panel	Task displayed correctly	Pass
Work Submission	File saved	File saved and timestamped	Pass
Final Delivery	Client downloads output	File downloaded successfully	Pass

10.15 Summary

This chapter presented the main outputs generated by the Nextwave digital agency system. All required functionalities show correct performance, demonstrating that the system is reliable, functional, and ready for further improvement in future phases.

Conclusion

11

11.1 Introduction

This chapter presents the final conclusion of the Nextwave digital agency system developed during the FYP-1 phase. It summarizes the project's purpose, achievements, and the key outcomes of the analysis, design, and documentation process. The conclusion highlights how the system fulfills its objectives and supports modern digital service operations.

11.2 Summary of the Project

Nextwave is designed as an online digital agency that provides multiple services such as web development, mobile applications, digital marketing, social media management, and graphic designing. The goal was to build a centralized platform where businesses can access professional services remotely without visiting a physical office.
Throughout the documentation, system requirements, architecture, workflows, use cases, and data flows were analyzed and defined in detail.

11.3 Achievement of Objectives

The project successfully achieved the following objectives:
• Clear system requirements were identified
• Functional and nonfunctional needs were defined
• Complete stakeholder analysis was conducted
• System architecture was designed with a scalable structure
• Data flow diagrams and use case models were developed
• Testing methods and implementation plans were documented
These achievements provide a strong technical foundation for FYP-2.

11.4 Importance of the System

Nextwave addresses a major problem faced by businesses, which is managing multiple digital service providers. The system simplifies this challenge by offering all required services in one

platform. Its remote nature reduces cost, improves communication, and allows easy access for clients from different regions.
The system is highly relevant in the modern digital era where businesses rely heavily on online solutions.

11.5 System Strengths

Key strengths of the Nextwave system include:
• Centralized service management
• Remote accessibility
• Organized workflow for service delivery
• Independent service modules
• Clear communication features
• Secure data handling and file management
These strengths improve overall efficiency and user experience.

11.6 Limitations Identified

Although the system is well structured, some limitations still exist:
• No real time chat system included in the initial version
• Automation is limited and depends on manual processes
• No AI based optimization in FYP-1
• Mobile app version is not yet implemented (planned for FYP-2)
These limitations provide direction for future enhancements.

11.7 Impact on Users

The system will benefit:
• Clients by offering quick digital services
• Businesses by providing reliable marketing and development support
• Students (project team) by gaining practical industry experience
• The university by showcasing a complete digital solution project
The system adds value on both professional and academic levels.

11.8 Learning Outcomes for the Team

The project helped the team develop:
• System analysis skills
• Documentation and research skills
• Problem solving abilities
• Understanding of digital agency operations
• Knowledge of modeling diagrams and workflows
• Enhanced teamwork and communication
These outcomes prepare the team for FYP-2 development.

11.9 Contribution to the Digital Domain

Nextwave contributes to the digital service domain by:
• Introducing an integrated service model
• Offering a complete remote-based agency structure
• Providing a scalable system design
• Addressing common industry challenges
The system's approach aligns with current digital transformation trends.

11.10 Final Closing Statement

Nextwave presents a strong foundation for building a fully functional digital agency platform. The FYP-1 documentation successfully covers all major aspects including requirements, architecture, design diagrams, workflows, and planning. With this foundation, the system is ready to enter the next phase, where it will be implemented and tested as a working solution in FYP-2.

11.11 Summary

This chapter concluded the documentation and highlighted the importance, achievements, strengths, limitations, and future potential of the Nextwave system. The project stands as a complete and well prepared digital solution for businesses seeking professional online services.

References

12

12.1 Introduction

This chapter provides all the references used during the research, analysis, and documentation of the Nextwave digital agency system. These sources include books, research articles, online documentation, and websites that support the concepts, techniques, and system analysis presented in this project. All references follow a simple and standard academic format.

12.2 Books and Academic Sources

1. Sommerville, I. (2016). *Software Engineering* 10th Edition. Pearson Education.
2. Pressman, R. S., & Maxim, B. R. (2014). *Software Engineering: A Practitioner's Approach.* McGraw Hill.
3. Satzinger, J., Jackson, R., & Burd, S. (2015). *Systems Analysis and Design in a Changing World.* Cengage Learning.
4. Dennis, A., Wixom, B., & Tegarden, D. (2019). *Systems Analysis and Design UML Version 2.0.* Wiley.

12.3 Online Articles and Documentation

1. "Understanding Data Flow Diagrams." Lucidchart. https://www.lucidchart.com
2. "UML Use Case Diagrams Explained." Visual Paradigm. https://www.visual-paradigm.com
3. "What is System Architecture?" IBM Documentation. https://www.ibm.com/docs
4. "Software Testing Fundamentals." Software Testing Help. https://www.softwaretestinghelp.com

12.4 Websites Related to Digital Agencies

1. HubSpot. "Digital Marketing Basics and Concepts." https://www.hubspot.com

2. Moz. “SEO and Digital Marketing Guides.” https://www.moz.com
3. Adobe Blog. “Creative Workflows and Design Trends.” https://www.adobe.com/blog
4. Google Developers. “Web Development Guides.” https://developers.google.com

12.5 Technology Documentation

1. React Official Documentation. https://react.dev
2. Node.js Official Docs. https://nodejs.org
3. MySQL Database Documentation. https://dev.mysql.com/doc
4. Figma Interface and Prototype Guide. https://help.figma.com

12.6 Research Resources Used for Domain Analysis

1. Statista. “Digital Transformation Market Growth.” https://www.statista.com
2. Forbes Tech. “Future of Digital Agencies.” https://www.forbes.com/technology
3. McKinsey Digital Report. “Business Digitalization Trends.” https://www.mckinsey.com